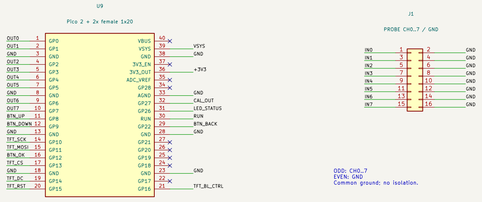
LOGIC ANALYZER — PICO 2 CARRIER / LAYOUT A

Rev A1 | 8 channels | 25 MS/s capture | USB-only | ST7735 SPI1 | engineering review

PICO MODULE / FEMALE SOCKETS



USB power/data through Pico only.
No external VSYS supply.
GP17..GP21, GP28 unconnected.
Keep BOOTSEL / module SWD accessible.

J1
PROBE CH0..7 / GND

ODD: CH0..7
EVEN: GND
Common ground: no isolation.

J2
TFT CABLE / 3V3 ONLY



1 GND 2 3V3 3 SCK 4 MOSI
5 RST 6 DC 7 CS 8 BL
Cable adapter to actual ST7735 module.

SW1
UP
BTN_UP
GND

SW2
DOWN
BTN_DOWN
GND

SW3
OK
BTN_OK
GND

SW4
BACK
BTN_BACK
GND

SW5
RESET
RUN
GND

Buttons: internal Pico pull-ups + firmware debounce.
RESET shorts RUN to GND; pull-up already on Pico.

C9
10u / 10V X7R
+3V3
GND

C10
10u / 10V X7R
+3V3
GND

J3
CAL OUT / GND
CAL_OUT 1
GND 2

CAL = GP27: connect to input with an external wire.

R41
1k
+3V3
LED0_A
D17
PWR
LED0_A
GND

R42
1k
LED_STATUS
LED1_A
D18
STATUS
LED1_A
GND

ASSEMBLY / VALIDATION NOTES

Two-layer carrier, 160 x 110 mm.
Pico 2 plugs into 2x20 female sockets.
TFT uses labeled cable connector J2.
No unverified direct-mating TFT footprint.

All eight Z nodes are independent.
Never connect Z/CLAMP to +3V3.
GPIO..7 connect only to buffer Y via 33R.

3V3 load target below 300mA: measure Pico + TFT + BL before release.

25 MS/s is sampling rate.
Input signal target 1kHz…1MHz: validate bandwidth and timing on board.

Engineering review, not production release.
High-voltage pulse rating is a design target; ESR/ORC cannot validate it.

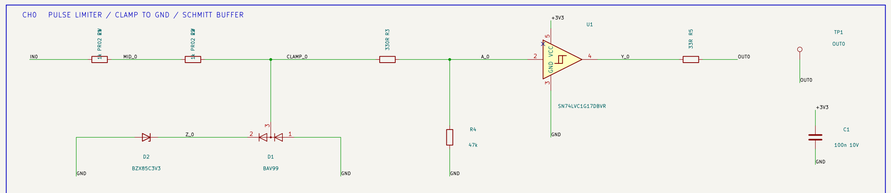
TFT BACKLIGHT — ACTIVE HIGH, DEFAULT 100R LIMIT



Default BL series 100R limits current <=33mA: can dim some modules.
Change only after checking module BL circuit/current. GP16 HIGH = ON.
Do not connect raw backlight LED to GP16.

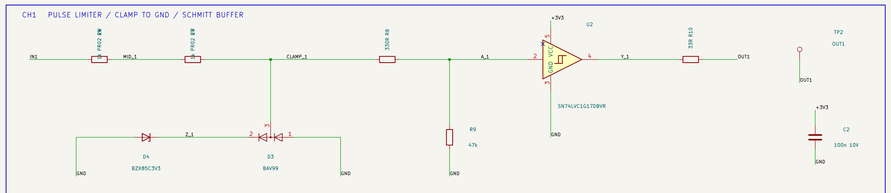
TP9 +3V3
TP10 VSYS
TP11 GND
TP12 IN0
TP13 CLAMP_0
TP14 A_0
TP15 Z_0

CH0 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



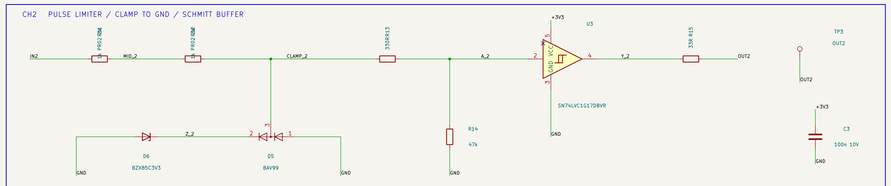
Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

CH1 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



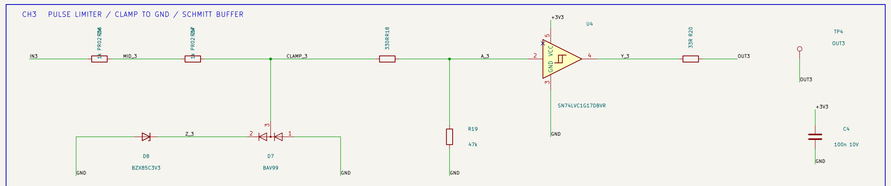
Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

CH2 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



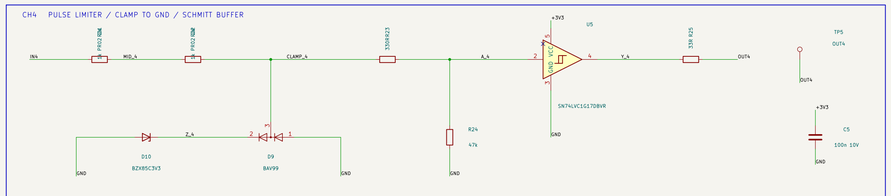
Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

CH3 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



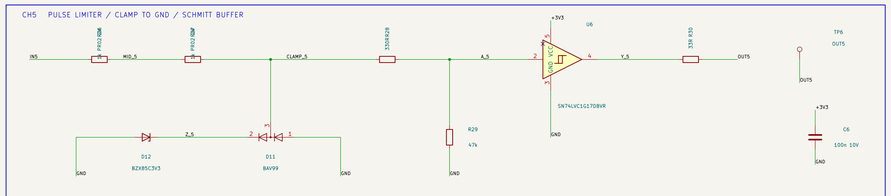
Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

CH4 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



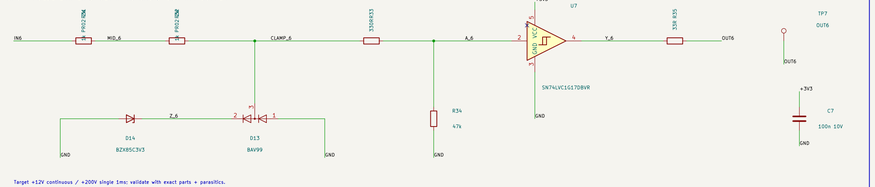
Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

CH5 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



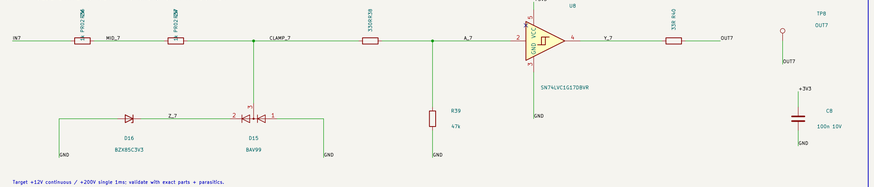
Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

CH6 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

CH7 PULSE LIMITER / CLAMP TO GND / SCHMITT BUFFER



Target +12V continuous / +200V single 1ms: validate with exact parts + parasitics.

H1 M3 / 3.2mm
H2 M3 / 3.2mm
H3 M3 / 3.2mm
H4 M3 / 3.2mm

Engineering review — see README for module mapping and validation limits
Logic Analyzer 2026.1
Sheet: /
File: LogicAnalyzer_Pico2.kicad_sch

Title: Pico 2 Logic Analyzer Carrier — Layout A

Size: A1	Date: 2026-11-11	**Rev: A1**
KiCad E.D.A. 10.0.8		Id: 1/1